



PRIMARY SCIENCE

TOPICAL REVISION

PROPERTIES OF MATTER



SUMMARIZED SCIENCE REVISION NOTES



PROPERTIES OF MATTER

Definition;

- Matter is anything that has weight and occupies space.
- Matter exists in three forms:- solids, liquids and gases .

a) Solids

- Things or objects we can hold with our hands.
- Particles are tightly packed.

b) Liquids

Particles are slightly loose and can move about.

- They pour and can only be carried using a container.

c) Gases

Particles are very loosely attached to each other.

The particles move freely in all directions. We cannot hold or touch gases.

FLOATING AND SINKING

- Solids which stay at the surface of water.
- Those materials which float on water are floaters.
- Those materials which sink in water are called sinkers.

Factors affecting sinking and floating

a) Types of the material

- Objects made of metal sink while those made of wood, rubber and plastic float.

b) Shape

- A bottle top which is not crushed floats on water.
- When the same bottle top is crushed, it sinks.

The crushed bottle top sinks because of the shape.

A ship made of iron and steel floats because of the shape.



- When the volume of an object increases, its density decreases, making it easier to float.

c) Weight of the material

- Materials which weigh more sink in water.
- When pebbles are added on a bottle top, it sinks because of increased weight..
- The size of material does not affect sinking and floating.

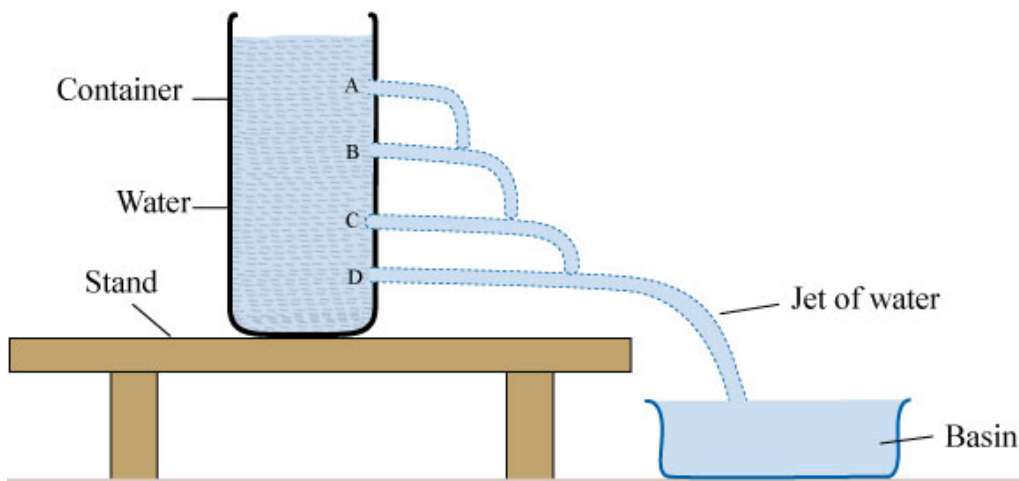
PRESSURE IN LIQUIDS

What is pressure?

It is a force that acts on a surface. Pressure comes from the word 'press' which means to apply force on something by pushing.

A liquid will only flow where there is pressure.

- Pressure in liquids increases with depth.
- Pressure is greatest at the bottom and least at the top.



- The hole at the bottom throws water at a greater distance since there is more pressure at the bottom than at the top.
- Pressure is equal at the same level or height.



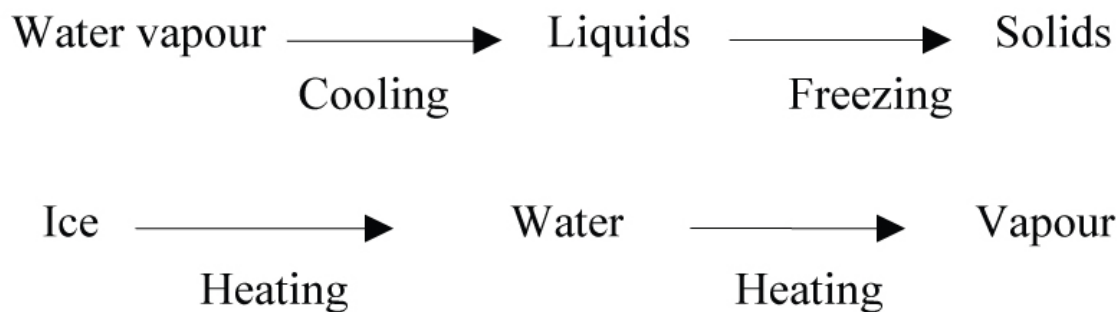
CHARACTERISTICS OF MATTER

SOLIDS	LIQUIDS	GASES
They have definite shape	Have no definite shape	Have no definite shape
They have definite mass	Have definite mass	Have definite mass
They have definite volume	Have definite volume or size	Have no definite volume

N.B. They all expand and contract on heating and cooling.

EFFECTS OF HEAT ON MATTER

Matter can change from one form to another when heat is decreased or increased e.g.



Matter when heated can melt, evaporate or expand, temperature also rises.

When heat is decreased, matter can condense, freeze or contract.

Melting

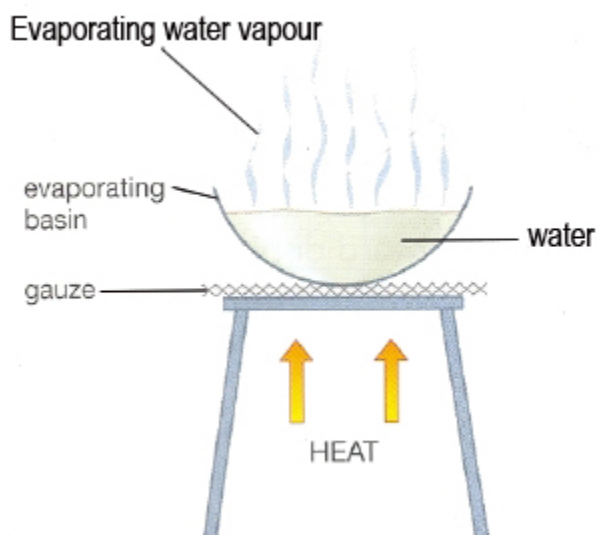
It is the changing of a solid e.g. ice into liquids

Evaporation

It is the changing of liquids into their vapours when heated e.g.

Water - Water vapour

Experiment - Evaporation of liquids.



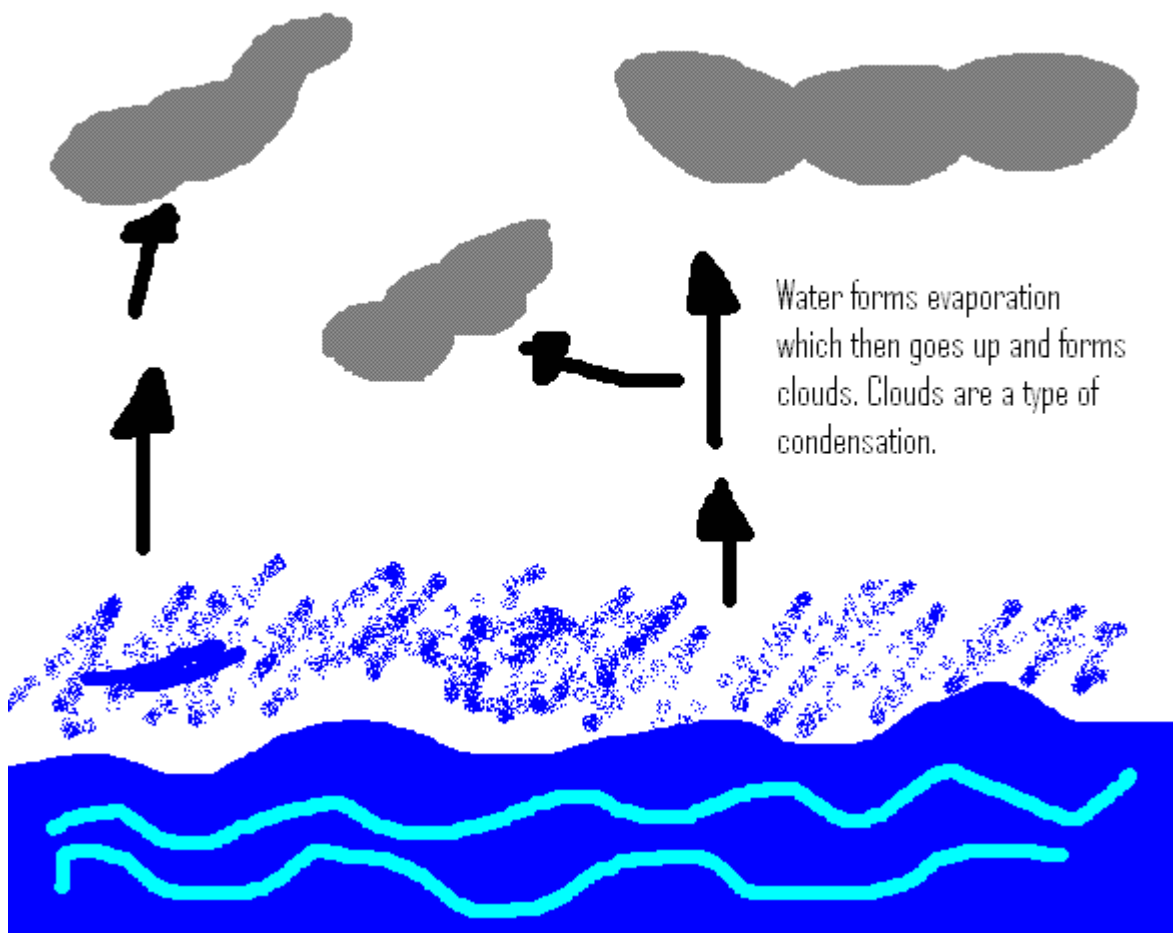
To get the correct conclusion

Amount of

- a) Liquids
 - b) Heat
 - c) Size of containers
 - d) Time taken to do the experiment must be the same.
- The liquids to be tested should not be the same.
 - Which one of the above liquids evaporates faster?
Spirit evaporated fastest.
 - Evaporation increases with increase in temperature

Condensation

- Changing of vapour into liquid.
- When water vapour or steam comes into contact with a cold surface, it changes back into liquids.



Freezing

- This is when liquids are cooled into solids.
- Different liquids freeze at different temperatures.
- Note the freezing point is the same as the melting point of a liquid.



EXPANSION AND CONTRACTION

When solids are heated, they expand and they contract when cooled.

a) Solids

- i) When the ball in the diagram is heated, it cannot pass through the ring.

This is because it has expanded.

- ii) Telephone and electric wires sag on a hot day due to expansion and tighten on a cold day due to contraction.



- iii) Gaps are left on railways lines to give room for expansion and prevent bending. Solids expand least in the three characteristics of matter.

b) Liquids

Water in a tightly-corked bottle rises up the straw due to expansion.

Some liquids expand more than others.

Liquids expand second highest.



c) Gases

- i) When the bottle with a balloon fitted on its neck is dipped in hot water, the balloon inflates.

This is because the air has expanded and occupied a bigger space.

- ii) Gases expand highest
- iii) When the air in the bottle shown is heated, bubbles of air escape through water. This is because heated air has expanded to occupy a larger space.

When heating is stopped, water rises up the straw.

Air inside the bottle cools down and contracts to occupy a smaller space, causing water to rise in the straw and occupy the space.

- Heat can also pass through a vacuum.

TEMPERATURE

It is the degree of hotness or coldness of a place, object or body.

- Temperature is measured using an instrument called a thermometer.
- Reading is done on a temperature scale in degrees Celsius (C).
- Temperature of ice is (0C) and pure boiling water is 100C.
- Mercury and alcohol are the commonly used liquids in thermometers.
- A clinical thermometer is used by doctors to measure body temperature.
- Thermometers work on the principle of expansion and contraction of liquids.

COMPOSITION OF AIR

GASES	PERCENTAGES
NITROGEN	78%
OXYGEN	21%
INERT GASES	0.97%
CARBON DIOXIDE	0.03%

Uses of Air

1. Oxygen

a) Breathing

- Oxygen is needed in our bodies to combine with food to produce energy.

b) Burning

- Oxygen is used in burning.
- Fuels e.g. charcoal, petrol cannot burn without oxygen.
- The burning candle in the diagram below has used oxygen in the glass forcing water to rise in the glass by a fifth.



c) Germination

- Seeds need air (oxygen), water and warmth to germinate.
- Seeds in boiled water covered with an oil layer did not germinate because they lacked oxygen.
- Water is boiled to expel air.



2. Carbon Dioxide

a) Making food for plants.

- Green plants make their own food through a process called photosynthesis.
- Sunlight, carbon dioxide, water and chlorophyll are needed during photosynthesis.

b) Preserving soft drinks

- It is added as a preservative during the manufacturing of soft drinks e.g. sodas

c) Making fire extinguishers

- Carbon dioxide and water are filled in a metallic can and used as a fire extinguisher.
- Carbon dioxide does not support burning.

3. Nitrogen

- Leguminous plants fix atmospheric nitrogen into the soil nitrogen using bacteria found in their root nodules.
- They use fixed nitrogen to make proteins.
- Legumes are sources of proteins e.g. peas, beans, groundnuts, Lucerne, desmodium

4. Inert gases

- They do not support burning.
- Argon is used in electric bulbs to prevent the filament from burning out.
- Neon is used in colored bulbs and tubes used in signboards and billboards.
- Helium is a light gas used in hot air balloons

SOLIDS THAT DISSOLVE IN WATER

Sugar and salt when mixed with water disappear after stirring.

Salt and sugar have dissolved in water.

When a solid dissolves completely in water, a solution is formed.



The solid e.g. sugar and salt that dissolves in a liquid is called a solute.

The liquid in which a solute dissolves is called a solvent e.g. water, milk etc.

Solute		Solvent		Solution
Sugar	+	Water	=	Sugar solution
Glucose	+	Water	=	Glucose Solution

SOLIDS THAT DO NOT DISSOLVE IN WATER

- Solids e.g. sand that do not dissolve in water are said to be insoluble.
- The sand settles at the bottom of the container.
- They do not make solutions.

Flour + Water = incomplete mixture

Sand + Water = incomplete mixture

Powdered milk + Water = incomplete mixture

Flour + Water = incomplete mixture

The solid that settles at the bottom of the insoluble mixture is called sediment.

MIXING LIQUIDS

Some liquids mix to form a uniform solution. This is called homogenous solution.

e.g. - Milk and water = homogenous solution

- Petrol and kerosene = homogenous solution
- Kerosene and cooking oil = homogenous solution
- Petrol and engine oil = homogenous solution

- Such liquids which form a homogenous solution are said to be miscible.

- Liquids which do not mix well are said to be immiscible liquids.
- Lighter liquids float on the heavier one. e.g. kerosene and water



- Kerosene floats on water.
- Liquids which do not mix are said to be heterogeneous.

MAGNETIC AND NON-MAGNETIC MATERIALS

Magnets

- Produce a force called magnetic force.
- Magnetic force attracts some objects. Such objects are magnetic materials.
- Those materials not attracted by a magnet are called non-magnetic.

Materials attracted by magnets	Materials not attracted by magnets
Tin	Aluminum
Iron	Silver
Nickel	Copper
Steel	Gold
Cobalt	Rubber
Chromium	Glass
Alnico	Plastic
	Brass

SEPARATING MIXTURES

Several methods are used depending on a mixture such as

- Soluble and insoluble
- Magnetic and non magnetic

Methods of separation include

- Winnowing
- Filtering
- Evaporation
- Sieving
- Decanting
- Picking
- Use of magnets

Methods of separating mixtures

a) Picking

- It is used to separate big solid particles e.g. mixture of beans and peas, peas and maize.

b) Winnowing

- It is used to separate light and heavy solids e.g. beans and husks, rice and husks.
- Husks are blown away the wind.

c) Sieving

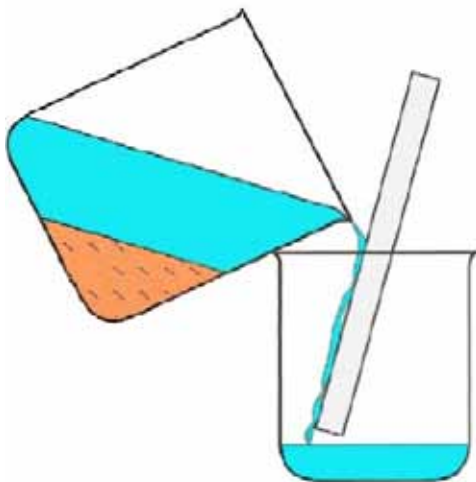
- It is used to separate small and large solid particles.
- Small particles pass through the sieve leaving large ones. e.g. maize, flour and rice.

d) Using a magnet

- It is used to separate a mixture of magnetic and non-magnetic materials e.g. iron fillings and flour.

e) Decanting

- It is a process of separating insoluble solids from a liquid by gently pouring out the liquid, e.g. Sand and water, beans and water.



- Immiscible liquids can also be decanted.

- f) Filtering - Insoluble solids are separated from a liquid using a piece of cloth or filter paper.



- g) Evaporation

- It a process used to separate a soluble solid from a liquid e.g. sugar and water.
- Water evaporates and salt remains in the evaporating container.
- In this separation method, only one substance is recovered.
- Solute is recovered while water (solvent) is lost to the atmosphere through evaporation.

- h) Filtering and Evaporation

- A mixture of water, sand and sugar.
- Sugar dissolves in the water but sand is insoluble.
- After the sand settled at the bottom, filter the mixture.
- Sugar solution is acquired while the sand remains in the filtering paper.
- Sugar solution is heated where sugar remains in the evaporating container.
 - Water evaporates into the atmosphere.
 - Sugar and sand are recovered but water is lost.